

Statistics

[illegible]

What Is Statistics?

1. **Collecting Data**
e.g., Survey
2. **Presenting Data**
e.g., Charts & Tables
3. **Characterizing Data**
e.g., Average

**Data
Analysis**

Why?



**Decision-
Making**



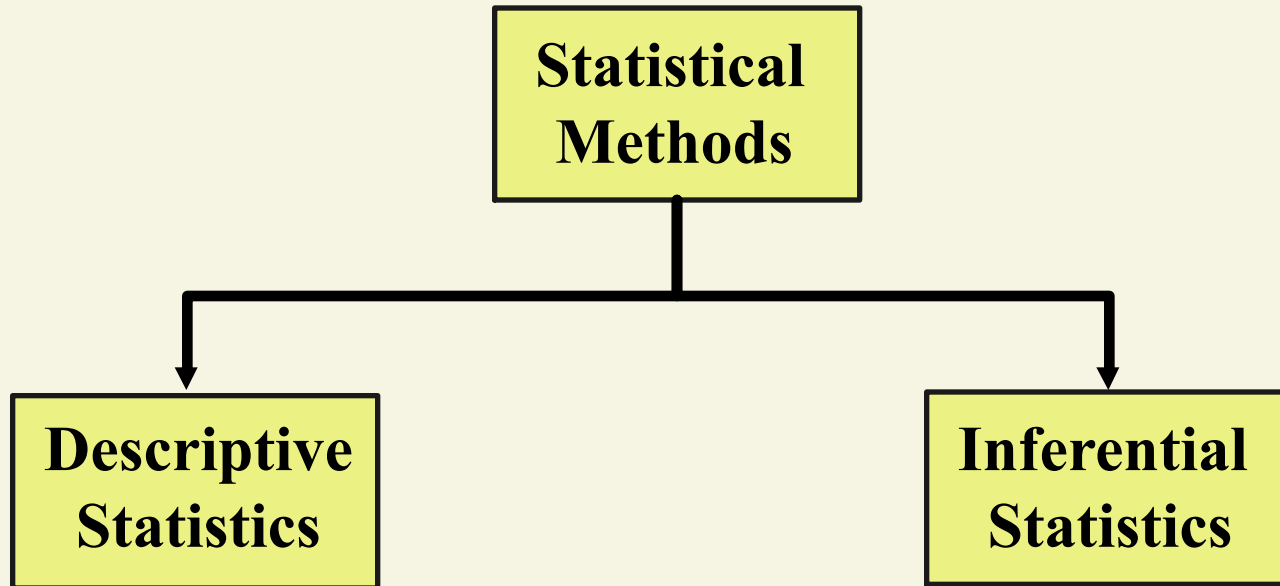
What Is Statistics?

- **Statistics** is the science of data.
- **Statistics** is the science of collecting, organizing, summarizing, and analyzing information to draw conclusions or answer questions. In addition, statistics is about providing a measure of confidence in any conclusions.

Statistics: Two Processes

- Describing sets of data and
- Drawing conclusions (making estimates, decisions, predictions, etc. about sets of data based on sampling)

Statistical Methods



Descriptive Statistics

- **Descriptive statistics** are methods for organizing and summarizing data.
- For example, tables or graphs are used to organize data, and descriptive values such as the average score are used to summarize data.

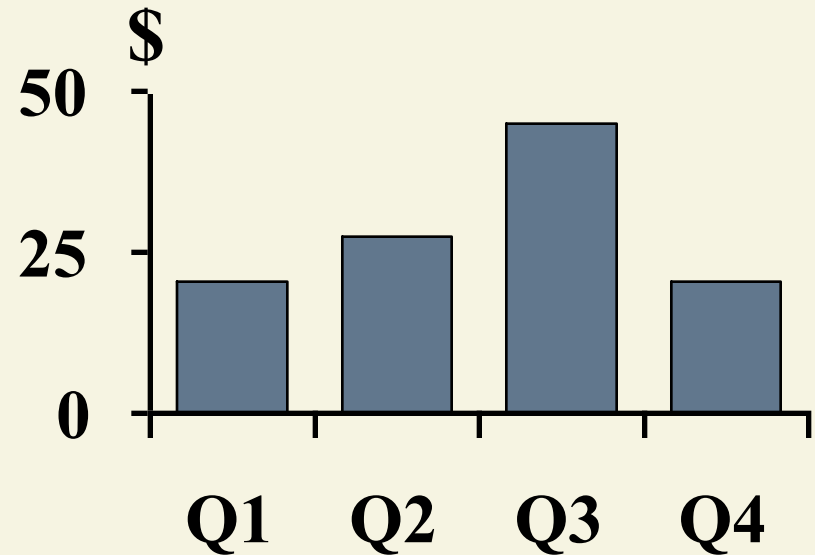
Descriptive Statistics

1. Involves

- Collecting Data
- Presenting Data
- Characterizing Data

2. Purpose

- Describe Data



$$\bar{X} = 30.5 \quad S^2 = 113$$

Inferential Statistics

- **Inferential statistics** are methods for using sample data to make general conclusions (inferences) about populations.

Inferential Statistics

1. Involves

- Estimation
- Hypothesis Testing

2. Purpose

- Make decisions about population characteristics



Population?



Four Elements of Descriptive Statistical Problems

1. The population or sample of interest
2. One or more variables (characteristics of the population or sample units) that are to be investigated
3. Tables, graphs, or numerical summary tools
4. Identification of patterns in the data

Five Elements of Inferential Statistical Problems

1. The population of interest
2. One or more variables (characteristics of the population units) that are to be investigated
3. The sample of population units
4. The inference about the population based on information contained in the sample
5. A measure of reliability for the inference

Fundamental Elements of Statistics

Population

- The entire group of individuals is called the **population**.
- We may be interested in the level of education in primary schools. All the children in the primary schools will make a population.

Population

- The population may contain living or non-living things.
- The entire lot of anything under study is called population.
- All the fruit trees in a garden, all the patients in a hospital and all the cattle in a cattle form are examples of population in different studies.

Finite and Infinite Populations

- Finite if it is of fixed and limited size
- Finite if it can be counted
- Infinite if it is unlimited
- Infinite if listing or counting every element is impossible

Examples of Finite Population

- The number of vehicles crossing a bridge every day, the number of births per years and the number of words in a book are finite populations.
- The number of units in a finite population is denoted by N . Thus N is the size of the population.

Examples of Infinite Population

- Let us suppose that we want to examine whether a coin is true or not. We shall toss it a very large number of times to observe the number of heads. All the tosses will make an infinite or countable infinite population.
- The number of germs in the body of a patient of malaria is perhaps something which is uncountable.

Sample

- Usually populations are so large that a researcher cannot examine the entire group.
- Therefore, a **sample** is selected to represent the population in a research study.
- The goal is to use the results obtained from the sample to help answer questions about the population.

Population and sample

